

LECTURE 03

MULTIVARIABLE CALCULUS

DIRECTION COSINES AND DIRECTION RATIOS OF A LINE JOINING

Two Points: For a line joining two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ the direction ratios are $x_2 - x_1, y_2 - y_1, z_2 - z_1$ and the direction cosines are $\frac{x_2 - x_1}{|PQ|}, \frac{y_2 - y_1}{|PQ|}, \frac{z_2 - z_1}{|PQ|}$.

Example: For a line joining two points $P(1, 3, 2)$ and $Q(7, -2, 3)$ the direction ratios are

$$7 - 1 = 6, -2 - 3 = -5, 3 - 2 = 1 \Rightarrow (6, -5, 1)$$
$$|PQ| = \sqrt{6^2 + (-5)^2 + 1^2}$$

And the direction cosines are

$$\left(\frac{6}{\sqrt{62}}, -\frac{5}{\sqrt{62}}, \frac{1}{\sqrt{62}} \right)$$

GRAPH IN TWO DIMENSIONAL SPACE

In two dimensional space the graph of an equation relating the variables x and y is the set of all point (x, y) whose coordinates satisfy the equations usually such graphs are curves.

Graph in Three Dimensional Space:

In three dimensional space the graph of an equation relating the variables x , y , and z is the set of all point (x, y, z) whose coordinates satisfy the equation, usually, such graphs are surfaces.

INTERSECTION OF TWO PLANES

If the two planes are not parallel, then they intersect and their intersection is a straight line. Thus, two non-parallel planes represent a straight line given by two simultaneous linear equations in x , y and z and are known as non-symmetric form of equations of a straight line.

Region	Description	Equation
xy-plane	Consists of all points of the form $(x, y, 0)$	$z = 0$
xz-plane	Consists of all points of the form $(x, 0, z)$	$y = 0$
yz-plane	Consists of all points of the form $(0, y, z)$	$x = 0$
x-axis	Consists of all points of the form $(x, 0, 0)$	$y = 0, z = 0$
y-axis	Consists of all points of the form $(0, y, 0)$	$x = 0, z = 0$
z-axis	Consists of all points of the form $(0, 0, z)$	$x = 0, y = 0$

GENERAL EQUATION OF PLANE (IN THREE DIMENSION)

Any equation of the form

$$ax + by + cz + d = 0$$

where, a, b , c , d are real numbers represent a plane.

Sphere:

The standard equation of the sphere radius “a” centered at (x_0, y_0, z_0) is

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2$$

EXAMPLE

Find center and radius of sphere

$$x^2 + y^2 + z^2 - 2x - 4y + 8z + 17 = 0$$

Solution: Completing the squares

$$(x^2 - 2x) + (y^2 - 4y) + (z^2 + 8z) + 17 = 0,$$

$$(x^2 - 2x + 1) + (y^2 - 4y + 4) + (z^2 + 8z + 16) = -17 + 21$$

$$(x - 1)^2 + (y - 2)^2 + (z + 4)^2 = 4$$

Radius = 2 Center: (1, 2, 4)

EXERCISE

Find the center and radius of the sphere

1) $x^2 + y^2 + z^2 - 2x - 6y - 8z + 1 = 0$

2) $x^2 + y^2 + z^2 + 10x + 4y + 2z - 19 = 0$

3) $x^2 + y^2 + z^2 - y = 0$

4) $2x^2 + 2y^2 + 2z^2 - 2x - 3y + 5z - 2 = 0$

5) $x^2 + y^2 + z^2 + 2x - 2y + 2z + 3 = 0$

EXERCISE

Find an equation of the sphere with the center **C** and radius **r**

1. $C(0,0); r = 8$

2. $C(-2,4,-1); r = 6$

3. $C(5,-2,4); r = \sqrt{7}$

4. $C(0,1,0); r = 3$

NATURAL DOMAIN

Natural domain consists of all points at which the formula has no divisions by zero and products only real numbers.

Example 01:

$$w = \sqrt{y - x^2}$$

The domain of $f(x, y) = \sqrt{y - x^2}$.

Consists of the shaded region and its bounding parabola $y = x^2$.

Domain: $y \geq x^2$

EXAMPLE 02

Domain of function $w = \frac{1}{xy}$ is the whole xy -plane excluding x -axis and y -axis.

Function: $w = \frac{1}{xy}$

Domain: $xy \neq 0$

Range: $(-\infty, 0) \cup (0, \infty)$

EXAMPLE 04

$$f(x, y) = xy\sqrt{y-1}$$

Domain of f consists of region in

xy - plane where $y \geq 1$

EXAMPLE 05

$$f(x, y) = \sqrt{x^2 + y^2 - 4}$$

Domain of f consists of region in

$xy - plane$ where $x^2 + y^2 \geq 4$

EXAMPLE 06

$$f(x, y) = \ln(xy)$$
$$xy > 0$$

Domain of f consists of region
lying in first and the 3rd
quadrants in xy -plane

EXAMPLE 07

$$f(x, y, z) = e^{xyz}$$

Domain of f consists of region of all three dimensional space.

EXAMPLE 08

$$f(x, y) = \frac{\sqrt{4 - x^2}}{y^2 + 3}$$

Domain of f consists of region in

$$\text{xy-plane } 4 - x^2 \geq 0,$$

$$x^2 \leq 4 \Rightarrow -2 \leq x \leq 2$$

EXAMPLE 09

$$f(x, y, z) = \sqrt{25 - x^2 - y^2 - z^2}$$

$$25 - x^2 - y^2 - z^2 \geq 0$$

$$x^2 + y^2 + z^2 \leq 25$$

Domain of f consists of region in three dimensional space occupied by sphere center at $(0, 0, 0)$ and radius 5.